Package 'DGrowthR'

April 9, 2025

Title Bacterial Growth Curve Analysis

Description A Comprehensive Toolset for Modelling of Bacterial Growth Curves via Gaussian Processes, Differential Testing and Empirical p-value Adjustment.

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Depends R (>= 2.10)

LazyData true

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Contents

`alternative` *Accessor for the alternative slot*

Description

This function retrieves the alternative slot from the provided DGrowthR object.

Usage

```
alternative(object)

## S4 method for signature 'DGrowthR'
alternative(object)
```

Arguments

object	An object from which to extract the alternative slot.

Value

The alternative slot of the object.

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calculate_emp_p_value *Calculate the Empirical p_value*

Description

This function calculates the Empirical p-value (emp_p_value) based on a given object.

Usage

```
calculate_emp_p_value(object)

## S4 method for signature 'DGrowthR'
calculate_emp_p_value(object)
```

Arguments

object	A DGrowthR object which contains the Gaussian Process regression results.

Value

A numeric value representing the calculated Empirical p-value.

---

clustering *Clustering Function*

Description

This function performs clustering using the specified algorithm (DBSCAN or GMM) on UMAP or fPCA coordinates.

Usage

```
clustering(object, ...)

## S4 method for signature 'DGrowthR'
clustering(
  object,
  embedding = "umap",
  algorithm = "dbscan",
  k = 0.01,
  eps = NULL
)
```

Arguments

`object`	A DGrowthR object containing preprocessed data including UMAP coordinates.
`...`	Additional parameters to clustering algorithm
`embedding`	A string indicating if the coordinates from "umap" or "fpca" should be used for clustering.
`algorithm`	A string specifying the clustering algorithm to use ("dbscan" or "gmm").
`k`	Numeric value indicating the number of nearest neighbors (for DBSCAN) or number of mixture components (for GMM). If $k < 1$, it is interpreted as a proportion of the total number of curves.
`eps`	Numeric value indicating the epsilon distance to use for DBSCAN clustering. If NULL, automatic selection based on k is performed.

Value

Updated DGrowthR object with updated metadata dataframe indicating cluster membership.

See Also

`dbscan::dbscan()`, `mclust::Mclust()`

`DGrowthR-class` *This S4 class represents a DGrowthR model.*

Description

It is used to store and validate data and parameters necessary for DGrowthR analysis.

Slots

- `od_data` A data.frame expected to contain the following columns: "timepoint", "od", and "curve_id". These represent the time points, and optical density (od), and an identifier for each curve respectively.
- `metadata` A data.frame that contains the "curve_id" column along with any metadata variables
- `raw_od` Optical density information before pre-processing or re-formatting.
- `verbose` A logical value indicating whether the output should be suppressed. Defaults to TRUE, i.e., output is not suppressed.

`fpca` An fpca object as the output of FPCA from fdapace

`umap_coord` Coordinates for UMAP embedding

`cluster_assignment` A data.frame matching curve_ids to the cluster assigned by clustering() function.

`gpfit_info` A list with the estimated mean gp process and the parameters used for fitting.

`growth_comparison` The results of a growth comparison.

`log_od` A logical indicating if the OD data has been logged.

`preprocessed` A logical variable indicating if the od data has been preprocessed with the preprocess_data() function.

`growth_parameters` A data.frame with the estimated growth parameters after gp modelling

Validity

For an object to be a valid DGrowthR, it must meet the following criteria:

- The "od_data" slot must be a data frame containing the columns "timepoint", "timepoint_n", "curve_id", and "od".
- The "metadata" slot must be a data frame with the "curve_id" column.
- All of the "curve_id"s in the "od_data" data frame must be present in the "curve_id" column of the metadata.

`DGrowthRFromData` *Instiantiate a DGrowthR object.*

Description

This function receives a dataframe with the optical density measurements, and, optionally and metadata data frame.

Usage

```
DGrowthRFromData(od_data, metadata = NULL, verbose = TRUE)
```

Arguments

`od_data`	A data.frame containing all of the optical density data. It should have a "Time" column with each row being a timepoint and the rest of the columns being an individual growth curve. The names of the growth curves should match those in the "curve_id" field in metadata.
`metadata`	Optional. A data.frame containing a "curve_id" field and all additional metadata. If none is provided, then one is created automatically containing only the wells from the curve_ids in od_data.
`verbose`	A logical variable indicating if DGrowthR object should be verbose.

Value

A DGrowthR object with filled raw_od, od_data, and metadata slots.

estimate_fpca *Functional Principal Component Analysis*

Description

This function estimates the functional principal components of the complete dataset using the fdapace package

Usage

```
estimate_fpca(object, data = "od_data", ...)

## S4 method for signature 'DGrowthR'
estimate_fpca(object, data = "od_data", ...)
```

Arguments

object	DGrowthR object.
data	A string indicating the source data to embed. At the moment only embeds "od_data"
...	Additional arguments to pass to the optns argument of FPCA

Value

The result from FPCA function from fdapace package

See Also

`fdapace::FPCA()`

estimate_growth_parameters *Estimate growth parameters*

Description

This function estimates a collection of growth parameters from the mean GP fits. If necessary, it also runs the GP fits.

Usage

```
estimate_growth_parameters(
  object,
  model_covariate = "curve_id",
  predict_n_steps = 100,
  downsample_every_n_timepoints = 1,
  sample_posterior_gpfit = FALSE,
  od_auc_at_t = NULL,
  sample_n_curves = 100,
```

```
  save_gp_data = FALSE,
  n_cores = 1
)

## S4 method for signature 'DGrowthR'
estimate_growth_parameters(
  object,
  model_covariate = "curve_id",
  predict_n_steps = 100,
  downsample_every_n_timepoints = 1,
  sample_posterior_gpfit = FALSE,
  od_auc_at_t = NULL,
  sample_n_curves = 100,
  save_gp_data = FALSE,
  n_cores = 1
)
```

Arguments

`object` A DGrowthR object containing preprocessed data.

`model_covariate`
: A string indicating a covariate in metadata to pool growth curves for GP modelling.

`predict_n_steps`
: A numeric value indicating the number of timepoints to sample from the mean posterior of the GP fit.

`downsample_every_n_timepoints`
: A numeric value indicating that the OD from every n timepoint should be used for GP fit. Might seep up fitting.

`sample_posterior_gpfit`
: A logical value. Indicates if the posterior GP should be sampled for new growth curves and parameters are estimated from these sampled curves. Allows to estimate mean and standard deviation of growth parameters. Really only makes sense if model_covariate pools more than one growth curve per group.

`od_auc_at_t` A numeric value indicating a specific timepoint for which the predicted OD and AUC should be returned.

`sample_n_curves`
: A numeric values. If sample_posterior_gpfit is TRUE, then sample_n_curves are sampled from posterior.

`save_gp_data` A logical value, indicating if the mean GP values and GP fit parameters should be saved to object.

`n_cores` A numeric values indicating the number of cores the user wants to use to model curves in parallel.

Value

Updated DGrowthR object with updated growth_parameters and, if requested, gpfit_info slots.

`estimate_umap` *Functional UMAP*

Description

This function estimates the UMAP coordinates using the uwot package

Usage

```
estimate_umap(object, data = "od_data", ...)

## S4 method for signature 'DGrowthR'
estimate_umap(object, data = "od_data", ...)
```

Arguments

`object`	DGrowthR object.
`data`	A string indicating the source data to embed. At the moment only embeds "od_data"
`...`	Additional arguments to pass to the optns argument of uwot

Value

The result from umap function from uwot package

See Also

`uwot::umap()`

`fit_predict_gp` *Fit at GP model and predict the mean process*

Description

This function fits a Gaussian Process model to a data.frame and then generates predictions from the fitted model.

Usage

```
fit_predict_gp(
  df,
  t_steps = 1000,
  complete_sigma = FALSE,
  prepare_dataframe = FALSE,
  estimate_derivatives = FALSE,
  delete = TRUE,
  pred_for_t = NULL
)
```

Arguments

- `df` A data.frame with the data that is to be modelled. Must have timepoint, timepoint_n, curve_id, and od fields.
- `t_steps` the number of timepoints within the range od df$timepoints to predict
- `complete_sigma` A logical indicating whether the complete covariance matrix should be returned. This is necessary if you want to re-sample the posterior.
- `prepare_dataframe` A logical indicating whether a data.frame with the mean data should be prepared.
- `estimate_derivatives` A logical indicating whether the empirical derivatives should be computed.
- `delete` A logical indicating whether the GP model should be deleted from memory FALSE.
- `pred_for_t` A numeric value inidicating a specific timepoint t for which OD should be predicted

Value

An object with the fitted model and prediction results. The results include the predictions (p) and the prediction inputs (XX).

---

gather_od_data *Gather specific growth data from the DGrowthR object*

Description

Extracts the requested growth curves according to the requested metadata fields and variables.

Usage

```
gather_od_data(
  object,
  metadata_field,
  field_value,
  downsample_every_n_timepoints = 1
)

## S4 method for signature 'DGrowthR'
gather_od_data(
  object,
  metadata_field,
  field_value,
  downsample_every_n_timepoints = 1
)
```

Arguments

`object`	An object of class "DGrowthR".
`metadata_field`	A character string specifying one of the fields in the metadata from which the variables will be searched in.
`field_value`	The value of the metadata_field that for which the relevant growth curves will be extracted.
`downsample_every_n_timepoints`	A numeric value indicating that the OD from every n timepoint should be used for GP fit. Might seep up fitting.

Value

A data.frame containing the optical density data requested.

`growth_comparison` *Predict the values of the fitted model to an object.*

Description

This function fits a Gaussian Process model to a DGrowthR object and then generates predictions from the fitted model. The function supports "alternative" and "null" models. The "alternative" model includes both time and contrast as predictors, while the "null" model only includes time as a predictor.

Usage

```
growth_comparison(
  object,
  comparison_info,
  predict_n_steps = 100,
  downsample_every_n_timepoints = 1,
  save_gp_data = FALSE,
  permutation_test = FALSE,
  n_permutations = NULL,
  n_cores = 1
)

## S4 method for signature 'DGrowthR'
growth_comparison(
  object,
  comparison_info,
  predict_n_steps = 100,
  downsample_every_n_timepoints = 1,
  save_gp_data = FALSE,
  permutation_test = FALSE,
  n_permutations = NULL,
  n_cores = 1
)
```

Arguments

`object` A DGrowthR object to which the model should be fitted and then predictions should be generated.

`comparison_info` character vector describing the contrast to be made. The first value indicates the column in metadata where the contrast takes place. The third value is taken as the reference condition.

`predict_n_steps` A numeric value indicating the number of timepoints to make a prediction for with the fitted GP model. More points increases estimate accuracy, but adds compute time.

`downsample_every_n_timepoints` A numeric value indicating that the OD from every n timepoint should be used for GP fit. Might speed up fitting.

`save_gp_data` A logical indicating whether the fitted GP models should be stored for future analysis.

`permutation_test` A logical value indicating if a permutation test should be performed.

`n_permutations` A numerical values indicating the number of permutations to build in order to gather the null distribution of test statistics.

`n_cores` A numeric value indicating the number cores to be used. Useful when performing a permutation test.

Value

An object with the updated growth_comparison slot.

---

growth_comparison_result

Accessor for the result of growth comparison

Description

This function retrieves the result of a growth comparison

Usage

```
growth_comparison_result(object)

## S4 method for signature 'DGrowthR'
growth_comparison_result(object)
```

Arguments

`object` An object from which to extract the perm_test_result slot.

Value

The growth comparison result data.frame

lrt_statistic *Accessor for the likelihood ratio test result*

Description

This function retrieves the likelihood ratio test result from the growth comparison slot

Usage

```
lrt_statistic(object)

## S4 method for signature 'DGrowthR'
lrt_statistic(object)
```

Arguments

object	An object from which to extract the perm_test_result slot.

Value

The LRT test statistic

multiple_comparisons *Multiple Comparisons*

Description

This function processes multiple contrasts, performs an analysis, and returns a data frame with the results.

Key features include:

- **Multiple contrast processing**: Ability to process and analyze multiple contrasts.
- **Comprehensive Analysis**: Performs a range of analyses based on user preferences, including Likelihood Ratio, empirical p-value, area under the curve (AUC) for both condition and control, maximum growth rate of the condition, maximum difference, and maximum optical density (OD) for both condition and control.
- **Output and Display Formats**: Provides an option to save results to a TSV file.

Usage

```
multiple_comparisons(
  object,
  comparison_list,
  predict_n_steps = 50,
  downsample_every_n_timepoints = 1,
  permutation_test = FALSE,
  n_permutations = NULL,
  n_cores = 1,
  write_to_tsv = FALSE,
  save_perm_stats = FALSE,
  filename = NULL
)
```

Arguments

`object` DGrowthR object that has been instantiated.

`comparison_list` A list of of vectors. Each vector contains the information for an individual contrast.

`predict_n_steps` A numeric value specifying the number of time points for prediction. Defaults to 50.

`downsample_every_n_timepoints` A numeric value that indicates how timepoints are sampled to increase fit speed

`permutation_test` A logical value indicating if a permutation test should be performed.

`n_permutations` Number of permutations for the permutation test (default: 100).

`n_cores` Number of cores to use for the permutation test (default: 1).

`write_to_tsv` A logical value inndicating if the results should be save to a tsv file.

`save_perm_stats` A logical value indicating whether the permuted test statistics should be stored in the DGrowthR object.

`filename` Name of the TSV file.

Value

Provides a data.frame summarizing the results.

`null` *Accessor for the null slot*

Description

This function retrieves the null slot from the provided DGrowthR object.

Usage

```
null(object)

## S4 method for signature 'DGrowthR'
null(object)
```

Arguments

`object` An object from which to extract the null slot.

Value

The null slot of the object.

perm_test *Perform a permutation test for the DGrowthR object*

Description

This function performs a permutation test for the DGrowthR object. It shuffles the "contrast" column in the OD data and calculates the difference in log-likelihoods between the alternative and null models for each permutation. The permutation test helps assess the significance of the difference in models.

Usage

```
perm_test(
  object,
  num_perms = 1000,
  n.cores = 1,
  predict_n_steps = 100,
  gp.delete = TRUE,
  ...
)

## S4 method for signature 'DGrowthR'
perm_test(
  object,
  num_perms = 1000,
  n.cores = 1,
  predict_n_steps = 100,
  gp.delete = TRUE,
  ...
)
```

Arguments

`object`	A DGrowthR object which contains the Gaussian Process regression results.
`num_perms`	The number of permutations to perform. Defaults to 1000.
`n.cores`	The number of CPU cores to use for parallel processing. Defaults to 1 (no parallel processing).
`predict_n_steps`	A numeric value indicating the number of timepoints to make a prediction for with the fitted GP model. More points increases estimate accuracy, but adds compute time.
`gp.delete`	A logical value indicating whether to delete the GPsep identifiers after the permutation test. Defaults to TRUE.
`...`	Additional arguments to be passed to the GP modelling function.

Value

A modified DGrowthR object with the calculated log-likelihood differences for each permutation.

`perm_test_result` *Accessor for the perm_test_result slot*

Description

This function retrieves the perm_test_result slot from the provided DGrowthR object.

Usage

```
perm_test_result(object)

## S4 method for signature 'DGrowthR'
perm_test_result(object)
```

Arguments

`object`	An object from which to extract the perm_test_result slot.

Value

The perm_test_result slot of the object.

`plot_fpca` *Plot FPCAs*

Description

This function plots the first two principal components calculated

Usage

```
plot_fpca(object, color = NULL)

## S4 method for signature 'DGrowthR'
plot_fpca(object, color = NULL)
```

Arguments

`object`	DGrowthR object.
`color`	Covariate by which to color the points in the scatterplot. Should be one of the columns in `metadata`. Defaults to NULL

Details

#—————————————————————————————————

Value

A scatterplot of the first two FPCs

plot_gp *Plot Gaussian Process*

Description

This function plots a specific gaussian process with the 95% confidence interval.

Usage

```
plot_gp(object, gpfit_id = NULL, show_input_od = FALSE)

## S4 method for signature 'DGrowthR'
plot_gp(object, gpfit_id = NULL, show_input_od = FALSE)
```

Arguments

`object`	An object of class "DGrowthR"
`gpfit_id`	The name of the GP model that should be plotted. Should be in the gpfit_info$gpfit_data
`show_input_od`	A logical value indicating if the od curves used to create the GP should be plotted

Value

A ggplot object showing lineplots of growth curves and GP mean.

plot_growth_comparison *Plot Growth Comparison*

Description

This function plots the null and the alternative fits to the model

Usage

```
plot_growth_comparison(object)

## S4 method for signature 'DGrowthR'
plot_growth_comparison(object)
```

Arguments

`object`	An object of class "DGrowthR"

Value

A list with two ggplot objects. One for the null model and another one for the alternative.

plot_growth_curves *Plot Optical Density*

Description

This function plots the raw optical density curves, with the possibility of specifying color and facet variables

Usage

```
plot_growth_curves(object, color = NULL, facet = NULL, data = "od_data")

## S4 method for signature 'DGrowthR'
plot_growth_curves(object, color = NULL, facet = NULL, data = "od_data")
```

Arguments

`object`	An object of class "DGrowthR"
`color`	A column of metadata to color growth curves
`facet`	A column of metadata to facet growth curves plots
`data`	A character value indicating which data should be plotted. If "raw_od" then unprocessed data is plotted. If "od_data" (default) then the pre-processed data is plotted if present.

Value

A ggplot object showing lineplots of growth curves.

plot_likelihood_ratio *Plot Likelihood Ratios*

Description

This function plots a histogram of permuted Likelihood Ratios with the actual Likelihood Ratio indicated by a vertical line.

Usage

```
plot_likelihood_ratio(object, num_bins = 50)

## S4 method for signature 'DGrowthR'
plot_likelihood_ratio(object, num_bins = 50)
```

Arguments

`object`	An object of class "DGrowthR".
`num_bins`	Number of bins to be used in the histogram.

Value

A ggplot object representing the histogram plot.

plot_od_difference *Plot Optical Density over Time*

Description

Calculates the optical density between the condition and control fitted curves for each time point, plots both models as well as the optical density difference between both fits.

Usage

```
plot_od_difference(object)

## S4 method for signature 'DGrowthR'
plot_od_difference(object)
```

Arguments

`object`	An object of class "DGrowthR".

Value

A ggplot object representing the line plot.

plot_umap *Plot UMAP components*

Description

This function plots the first two UMAP components calculated

Usage

```
plot_umap(object, color = NULL)

## S4 method for signature 'DGrowthR'
plot_umap(object, color = NULL)
```

Arguments

`object`	Degrowth object.
`color`	Covariate by which to color the points in the scatterplot. Should be one of the columns in `metadata`. Defaults to NULL

Value

A scatterplot of the first two UMAP components

preprocess_data *Preprocess Growth Data*

Description

This function reads growth data from a file, applies several optional preprocessing steps, and saves the processed data.

Usage

```
preprocess_data(
  object,
  baseline = 1,
  log_transform = TRUE,
  skip_first_n_timepoints = 0
)

## S4 method for signature 'DGrowthR'
preprocess_data(
  object,
  baseline = 1,
  log_transform = TRUE,
  skip_first_n_timepoints = 0
)
```

Arguments

`object` A DGrowthR object containing the growth data.

`baseline` A numeric value specifying timepoint that will be used for baseline correction. If a vector is provided, then the minimum OD of the provided timepoints is used as baseline.

`log_transform` A logical value indicating whether to perform log transformation on optical density measurements. This enables accurate estimation of growth parameters.

`skip_first_n_timepoints` A numeric value the number of timepoints to skip for all growth curves. The analysis would start at skip_first_n_timepoints + 1.

Value

The processed DGrowthR object with updated od_data.

read_individual_plate *Read individual growth plate*

Description

This function receives a file name that contains plate reader OD data. It returns a data.frame with the parsed data.

Usage

```
read_individual_plate(plate_file, path_target, time_vector = c(), sep = "\t")
```

Arguments

`plate_file`	A character variable that is the file name for the plate OD data. Should contain a Time column and a column for each well.
`path_target`	The path to the plate file
`time_vector`	A numeric vector that indicates the Time at which each measurement was taken.
`sep`	The delimiter character that separates columns in the plate file

Value

A data.frame with the data from the plate file. The Time column contains the values in the time_vector, and the wells are renamed to be in format `{plate_name}_{well}`.

`read_multiple_plates` *Read a collection growth plate data.*

Description

This function receives a path to a folder that contains data for several plates. Each plate should have its own file.

Usage

```
read_multiple_plates(path_target, time_vector = c(), sep = "\t")
```

Arguments

`path_target`	The path to the folder containing plate files
`time_vector`	A numeric vector that indicates the Time at which each measurement was taken. If empty then the Time column from one of the plates is taken as reference.
`sep`	The delimiter character that separates columns of each file in path_target

Value

A list of two dataframes. An "od_data" dataframe with a common time column and each growth curve in a separate column named with its "curve_id" in format `{plate_name}_{well}`. Also a "metadata" dataframe indicating the well and the original plate from which each "curve_id" is gathered from.

`show,DGrowthR-method` *Display a summary of the DGrowthR object*

Description

This method displays a summary of the DGrowthR object, including the first few rows of the od_data slot and the log-likelihood and posterior probabilities of the alternative and null models.

Usage

```
## S4 method for signature 'DGrowthR'
show(object)
```

Arguments

`object`	A DGrowthR object.